

ASSIGNMENT 2

• SESSION 12

1. Write equivalence partitioning (EP) test cases for the phone number input field in a Zomato-style food order form, where the field must accept exactly 10 digits and no letters or special characters.

Ans.

Requirement :-

- Must accept **exactly 10 digits**
- No letters or special characters allowed

Equivalence Classes :-

Partition Description		Valid/Invalid
EC1	Exactly 10 numeric digits	Valid
EC2	Less than 10 digits	Invalid
EC3	More than 10 digits	Invalid
EC4	Contains letters	Invalid
EC5	Contains special characters/spaces	Invalid
EC6	Empty input	Invalid

Test Cases :-

TC ID	Test Input	Expected Result	Partition
TC1	9876543210	Accepted	EC1
TC2	987654321	Error: Must be 10 digits	EC2
TC3	98765432101	Error: Must be 10 digits	EC3
TC4	98765abc10	Error: Only digits allowed	EC4
TC5	98765@3210	Error: Special characters not allowed	EC5
TC6	(blank)	Error: Phone number required	EC6

2. For a Flipkart-style sign-up form with a password field that must be 8-16 characters long, write boundary value analysis (BVA) test cases to check the minimum, just above minimum, maximum, and just above maximum password lengths.

Ans.

Requirement :-

- Password length: **8–16 characters**

BVA Test Cases :-

TC ID	Password Length	Example	Expected Result
TC1	7	Abc1234	Rejected
TC2	8 (Minimum)	Abc12345	Accepted
TC3	9 (Just Above Minimum)	Abc123456	Accepted
TC4	15	Abc123456789012	Accepted
TC5	16 (Maximum)	Abc1234567890123	Accepted
TC6	17 (Just Above Maximum)	Abc12345678901234	Rejected

3. Given an Instagram-like post creation form with a caption input that allows up to 2200 characters, use equivalence partitioning to identify valid and invalid input ranges and write at least 3 test cases for each partition.

Ans.

Requirement :-

- Caption length: **0–2200 characters**

Equivalence Classes :-

Partition	Description	Valid/Invalid
EC1	0–2200 characters	Valid
EC2	More than 2200 characters	Invalid

Valid Partition Test Cases (EC1) :-

TC ID	Input	Expected Result
TC1	Empty caption	Accepted
TC2	100-character caption	Accepted
TC3	2200-character caption	Accepted

Invalid Partition Test Cases (EC2) :-

TC ID	Input	Expected Result
TC4	2201-character caption	Error displayed
TC5	2300-character caption	Error displayed
TC6	5000-character caption	Error displayed

4. For a BookMyShow movie ticket booking form with a 'number of tickets' field (allowed: 1 to 10), write BVA test cases to ensure the field correctly handles edge values.

Ans.

Requirement :-

- Allowed values: **1–10 tickets**

BVA Test Cases :-

TC ID	Number of Tickets	Expected Result
TC1	0	Rejected
TC2	1 (Minimum)	Accepted
TC3	2 (Just Above Minimum)	Accepted
TC4	9 (Just Below Maximum)	Accepted
TC5	10 (Maximum)	Accepted

TC ID	Number of Tickets	Expected Result
TC6	11	Rejected

5. Pick any app you use daily (such as Paytm, Myntra, or IRCTC) and select one form field (e.g., amount, pincode, age). Write both equivalence partitioning and boundary value analysis test cases for that field, explaining your reasoning for each.

Ans.

Field Chosen:- Amount

Requirement:-

- Amount allowed: ₹1 to ₹10,000

A. Equivalence Partitioning (EP)

Equivalence Classes

Partition Description	Valid/Invalid
EC1 ₹1–₹10,000	Valid
EC2 Less than ₹1	Invalid
EC3 Greater than ₹10,000	Invalid
EC4 Empty input	Invalid

EP Test Cases :-

TC ID	Input	Expected Result	Reason
TC1	₹500	Accepted	Valid amount
TC2	₹0	Rejected	Below minimum
TC3	₹10,001	Rejected	Above maximum
TC5	Blank	Rejected	Required field

Reasoning :-

- One representative value from each equivalence class is sufficient because all values within the same class are expected to behave similarly.

B. Boundary Value Analysis (BVA)

Boundary Values :-

TC ID	Amount	Expected Result	Reason
TC1	₹0	Rejected	Just below minimum
TC2	₹1	Accepted	Minimum boundary
TC3	₹2	Accepted	Just above minimum
TC4	₹9,999	Accepted	Just below maximum
TC5	₹10,000	Accepted	Maximum boundary
TC6	₹10,001	Rejected	Just above maximum

Reasoning

- BVA focuses on values at and around the boundaries because defects commonly occur at edge conditions. Testing the minimum, just above minimum, maximum, and just above maximum helps verify correct validation behavior.